

Alset IoT HTL

Team Smart Lane Savvy
Team 1

Team Members: Octavio Morales, Allan Joseph,
Anne-Marie Akoh, Joseph Bonetti





Our Goal

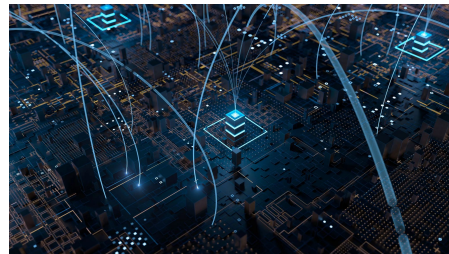
- Demonstrate the inherent capabilities of IoT
 - Flexibility, malleability, resourcefulness, extensive and in-depth range of information
- Improve the experiences of the average motorist
 - Lower inhibitions while driving - automatic safety mechanisms in place
 - Be more energy efficient for the environment
 - Ultimately improve our understanding of our technological capacity
- Minimise the cost of vehicle production
 - Main financial investments goes into developing the software and funding technician updates
 - Aside from that, not much more costly per car
- Communication with the Cloud
 - Company cloud can receive and analyse data from the car's software and Control System
 - Allows us to make informed decisions and draw conclusions regarding the habits/tendencies of drivers and road safety
 - Correlations can help us predict future changes to prepare software updates for



What's the Use? The Value?

Why should we even care to do this?

- There's value in development - gives us insight
- Imagine the benefits of a car-efficient economy
 - Less concern with carbon emissions
 - More time and money to dedicate towards other environmental issues
- And the benefits of safer driving in general
 - Increased road safety leads to
 - Fewer annual accidents
 - Decreased worry over car insurance
 - Higher motorist and pedestrian morale
 - (fewer accidents, safer roads, safer communities, potentially happier lives)
- Opening doors to more advanced innovation
 - If we can be successful in cars then what other opportunities does this open us up to?
 - Airplanes, trains, public transport in general...
 - Endless possibilities





IoT HTL Requirements

- Motion and No Motion
 - Battery Level Sensor, Motor Sensor, Brake Sensor, GPS sensor, Tire Pressure Sensor, Camera System
- Steering
 - GPS Navigation, Radio Wave Sensor
- Braking
 - Sonar Sensor, Camera System, Three-Dimensional Scanning Cameras
- Obstacle Detection
 - Sonar Sensor, Radio Wave Sensor, Camera System
- Navigation
 - Global Positioning System (GPS)





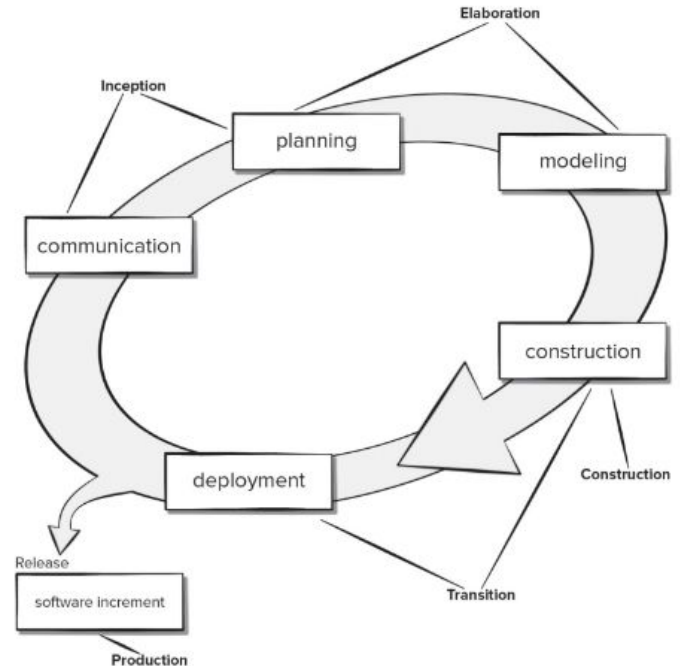
IoT HTL Requirements

- Acceleration
 - Sensor Fusion
- Traffic Sound Detection
 - Three-Dimensional Scanning Cameras
- Sound Detection
 - Three-Dimensional Scanning Cameras, Sound Sensors
- Crash Detection
 - Sonar Sensor
- Cruise Control
 - Sonar Sensor





- **Process Method**
 - Using the Unified Process method, our team hopes to manufacture high-level software that can simulate that of a self-driving car.
- **Benefits**
 - Allows Customer Involvement, Accommodates Requirement Changes and Maintenance Projects
- **Initial Process Development**
 - For the first release we plan to offer a first prototype of sensor detection. After this first release, revisions will be performed and edits will be made to improve upon our work. Then, subsequent releases will build upon their prior ones and follow suit. This can then be applied to every part.
- **Results**
 - With a project like this, it allows us to create the safest experience possible for all users.



Challenges and Excellences

- Challenges
 - Complexity: A driverless car has very complex hardware and software systems including cameras, sensors that all need to work together seamlessly to ensure safe and reliable operation
 - Integration: Once a model has been developed, it needs to be integrated into existing software systems or processes
 - Time and Cost: Developing a driverless car can be time consuming and expensive requiring significant investment in research, development, testing, and validation
- Excellences
 - Convenience: With a driverless car passengers can have a more relaxed and productive experience along with the ability for anyone without a license to drive
 - Efficiency: Driverless cars can optimize their routes and speed reducing traffic congestion and time of travel
 - Safety: Driverless cars have the potential to significantly reduce the number of traffic accidents caused by human error
- Cost Effective and Faster Development
 - Prioritize Modularity/Standardization: This would make it easier to integrate different components, reduce complexity, and streamline the development process
 - Outsourcing: Allows for there to be more focus on core tasks while non-core tasks can be done elsewhere
 - Agile Development: This would allow for smaller, high focus development cycles that allow for more rapid prototyping and testing
 - Open Source Software: Will reduce development costs by using pre existing software components that are openly available and customizable
 - Simulation Tools: Allows for faster testing and provides validation of certain scenarios



What Aspect of This Project Should Be Prioritized To Ensure The Safest Experience For All Users?



Final Remarks